

Fixed-Alpha Variational and PINN Formulations for the Reduced Radial Square-Well Problem

1 Mathematical Problem and Reference Solution

We consider the low-energy s -wave reduction of the deuteron relative-coordinate Schrödinger equation. With the reduced radial wavefunction

$$u(r) = r\psi(r), \quad (1)$$

the equation on the truncated finite interval $[0, r_{\max}]$ is

$$-Ku''(r) + V(r)u(r) = Eu(r), \quad 0 \leq r \leq r_{\max}, \quad (2)$$

where

$$K = \frac{\hbar^2}{2m}. \quad (3)$$

The physical constants are

$$m_p = 938.27 \text{ MeV}, \quad m_n = 939.57 \text{ MeV}, \quad (4)$$

$$m = \frac{m_p m_n}{m_p + m_n} \approx 469.460 \text{ MeV}, \quad \hbar = 197.327 \text{ MeV fm}. \quad (5)$$

Thus

$$K \approx 41.4710 \text{ MeV fm}^2. \quad (6)$$

The square-well potential is

$$V(r) = \begin{cases} -V_0, & 0 \leq r \leq R, \\ 0, & r > R, \end{cases} \quad V_0 = 35 \text{ MeV}, \quad R = 2.13 \text{ fm}. \quad (7)$$

The finite computational box is

$$r_{\max} = 20 \text{ fm}, \quad (8)$$

with boundary conditions

$$u(0) = 0, \quad u(r_{\max}) = 0, \quad (9)$$

and normalization

$$\int_0^{r_{\max}} |u(r)|^2 dr = 1. \quad (10)$$

For a bound-state energy $E < 0$, define

$$k = \sqrt{\frac{E + V_0}{K}}, \quad \kappa = \sqrt{\frac{-E}{K}}. \quad (11)$$

The finite-box square-well solution has the form

$$u_{\text{box}}(r) = \begin{cases} A \sin(kr), & 0 \leq r \leq R, \\ C \sinh(\kappa(r_{\text{max}} - r)), & R \leq r \leq r_{\text{max}}. \end{cases} \quad (12)$$

Continuity of u and u' at $r = R$ gives the finite-box eigenvalue equation

$$k \cot(kR) = -\kappa \coth(\kappa(r_{\text{max}} - R)). \quad (13)$$

For the parameters above, the reference finite-box energy is

$$E_{\text{box}} = -3.038467 \text{ MeV}. \quad (14)$$

This exact finite-box solution is used as the numerical reference for the error measurements below.

2 Variational Formulation and PINN Formulation

The variational formulation minimizes the Rayleigh quotient

$$\mathcal{E}[u] = \frac{\int_0^{r_{\text{max}}} (K|u'(r)|^2 + V(r)|u(r)|^2) dr}{\int_0^{r_{\text{max}}} |u(r)|^2 dr}. \quad (15)$$

For the ground state, minimizing $\mathcal{E}[u]$ under the boundary and normalization constraints gives the lowest eigenpair of the finite-box Hamiltonian.

The PINN formulation instead penalizes the local differential-equation residual. For a neural approximation u_θ and trainable energy E_θ , define

$$\mathcal{R}_\theta(r) = -K u_\theta''(r) + V(r)u_\theta(r) - E_\theta u_\theta(r). \quad (16)$$

A pure PINN objective is

$$\mathcal{L}_{\text{PINN}} = \lambda_{\text{pde}} \langle |\mathcal{R}_\theta|^2 \rangle + \lambda_{\text{norm}} \left(\int_0^{r_{\text{max}}} |u_\theta(r)|^2 dr - 1 \right)^2 + \lambda_D |u_\theta(r_{\text{max}})|^2. \quad (17)$$

The normalization term prevents the trivial zero solution, and the boundary term weakly enforces the finite-box condition at r_{max} .

For all neural experiments in this report, we use the fixed-alpha ansatz

$$u_\theta(r) = r e^{-\alpha r} N_\theta(r), \quad \alpha = 0.25 \text{ fm}^{-1}. \quad (18)$$

The factor r enforces $u_\theta(0) = 0$, and the exponential factor encourages a decaying bound-state tail. In the PINN-based losses, the energy is represented by

$$E_\theta = -\text{softplus}(E_{\text{raw}}), \quad (19)$$

so that the trainable energy remains negative.

The combined variational plus PINN formulation is

$$\begin{aligned} \mathcal{L}_{\text{var+PINN}} = & \lambda_{\text{ray}} \mathcal{E}[u_\theta] + \lambda_{\text{pde}} \langle |\mathcal{R}_\theta|^2 \rangle + \lambda_{\text{match}} (E_\theta - \mathcal{E}[u_\theta])^2 \\ & + \lambda_{\text{norm}} \left(\int_0^{r_{\text{max}}} |u_\theta(r)|^2 dr - 1 \right)^2 + \lambda_D |u_\theta(r_{\text{max}})|^2. \end{aligned} \quad (20)$$

The Rayleigh term is ground-state seeking; the PDE residual enforces local equation consistency; and the energy-consistency penalty couples the learned eigenvalue to the variational energy estimate.

3 Numerical Comparison of Three Strategies

We compare the following strategies:

1. Variational formulation only:

$$\mathcal{L}_{\text{var}} = \mathcal{E}[u_\theta] + 10 (\|u_\theta\|_2^2 - 1)^2 + 10 |u_\theta(r_{\text{max}})|^2. \quad (21)$$

2. PINN only:

$$\mathcal{L}_{\text{PINN}} = \langle |\mathcal{R}_\theta|^2 \rangle + 10 (\|u_\theta\|_2^2 - 1)^2 + 10 |u_\theta(r_{\text{max}})|^2. \quad (22)$$

3. Variational formulation plus PINN:

$$\mathcal{L}_{\text{var+PINN}} = \mathcal{E}[u_\theta] + \lambda_{\text{pde}}(t) \langle |\mathcal{R}_\theta|^2 \rangle + \lambda_{\text{match}}(t) (E_\theta - \mathcal{E}[u_\theta])^2 + 10 (\|u_\theta\|_2^2 - 1)^2 + 10 |u_\theta(r_{\text{max}})|^2. \quad (23)$$

For the combined strategy, the PDE and energy-consistency weights are selected by a gradient-based learning schedule. The Rayleigh term is used as the anchor:

$$\lambda_i(t) \approx c_i \frac{\|\nabla_\theta \mathcal{E}[u_\theta]\|}{\|\nabla_\theta \mathcal{L}_i\| + \epsilon}, \quad \mathcal{L}_i \in \{ \langle |\mathcal{R}_\theta|^2 \rangle, (E_\theta - \mathcal{E}[u_\theta])^2 \}. \quad (24)$$

The training uses a 2000-step Rayleigh warmup, followed by a 3000-step ramp to the caps

$$\lambda_{\text{pde}} \leq 0.1, \quad \lambda_{\text{match}} \leq 0.5. \quad (25)$$

The neural network has 3 hidden layers, width 64, and tanh activations. All runs use SOAP with learning rate 3×10^{-3} for 10000 CUDA steps. The PDE residual uses 2240 random mixture samples per step: 1280 interior points, 320 near-origin points, and 640 interface points near $r = R$. The Rayleigh quotient and normalization use 2048 uniform random samples per step. Validation is recomputed on a fixed 8192-point grid.

Strategy	E_{val} [MeV]	$ E_{\text{val}} - E_{\text{box}} $ [MeV]	$\ u - u_{\text{box}}\ _2$	PDE loss	$u(20)$
Variational only	-3.029412	0.009055	0.018313	1.045433	4.56×10^{-3}
PINN only	1.345365	4.383832	1.244789	0.157523	1.43×10^{-2}
Variational + PINN	-3.036766	0.001701	0.024754	0.216411	4.73×10^{-3}

Table 1: Validation metrics for the three fixed- $\alpha = 0.25$ strategies after 10000 training steps. E_{val} is the Rayleigh quotient recomputed on a fixed 8192-point grid.

The variational-only formulation gives a good wavefunction and a good energy because the Rayleigh quotient directly targets the lowest finite-box eigenvalue. The PINN-only formulation achieves a relatively small local residual but does not recover the desired eigenpair: the learned energy collapses close to zero and the Rayleigh energy of the resulting wavefunction is positive. This illustrates that a residual loss by itself is not reliably ground-state seeking in an eigenvalue problem, even when normalization is included.

The combined variational plus PINN formulation gives the most accurate energy in this experiment. The gradient-based schedule prevents the residual and energy-consistency penalties from dominating before the Rayleigh objective has found the bound-state basin. However, its wavefunction L^2 error is slightly larger than the variational-only result, so the combined method should be viewed as an energy-improving and equation-regularizing strategy rather than an automatic improvement for every metric.

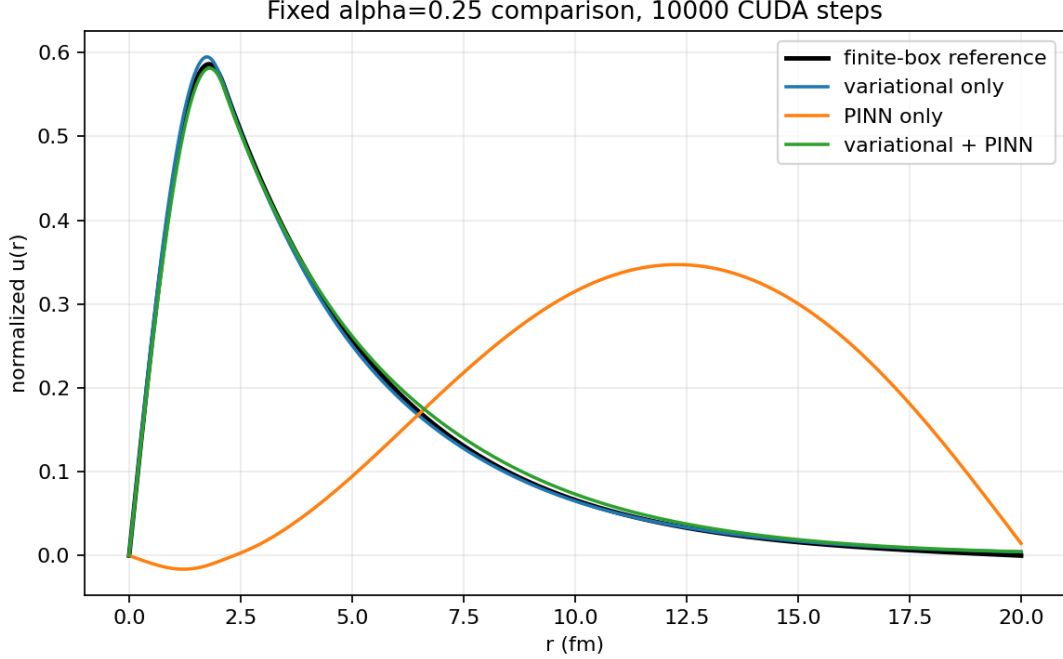


Figure 1: Normalized reduced radial solutions from the three fixed- $\alpha = 0.25$ strategies compared with the exact finite-box reference solution.

4 Caveats of the Variational Plus PINN Strategy

The combined variational plus PINN loss is attractive because it merges a global ground-state principle with local differential-equation information. It also has several caveats.

First, the loss terms have different units and different gradient scales. The Rayleigh term has units of energy, while the residual term is a squared equation error. Fixed weights can easily make the residual dominate the optimization, causing the network to satisfy a local equation around an incorrect learned energy. This was observed in earlier fixed-weight experiments.

Second, the learned energy E_θ and the Rayleigh energy $\mathcal{E}[u_\theta]$ need not agree unless the consistency term is weighted appropriately. If λ_{match} is too large too early, the network can be pulled toward a noisy or shallow energy estimate. If it is too small, E_θ may not represent the variational energy of the learned wavefunction.

Third, the fixed exponential ansatz parameter α constrains the tail shape. The value $\alpha = 0.25 \text{ fm}^{-1}$ is close to the finite-box reference decay rate but is still a modeling choice. For more general potentials or unknown energies, a fixed α may bias the solution and should be tested or made adaptive.

Finally, the square-well discontinuity creates a sampling-sensitive interface region. The PDE residual is most informative only if collocation points sufficiently resolve the interface. Without deliberate interface sampling, the residual loss can underrepresent the most difficult part of the solution.